Decent Coaching Classes

VIII CBSE

Mind Your Math - IX

ANSWER KEY & SOLUTIONS

Section A — MCQ (1 mark each) **[6]**

(1) C (2) B (3) A (4) B (5) D (6) B

Hints: (1) $7ab - 5a$ has exactly two terms. (2) $3x^2y$ and $-5yx^2$ have the same variables with the same powers. (3) $2x^2 - 8x + 3x - 12 = 2x^2 - 5x - 12$.

(4) $4x^2 - 25 = (2x)^2 - 5^2$. (5) $5x - 10 = 3x + 6$, so $2x = 16$, $x = 8$. (6) $x^2 + kx + 36 = (x \pm 6)^2$ needs $k = 2 \times 6 = 12$.

Section B — (2 marks each) **[8]**

7. Coefficient of x^2y^2 = **5**; constant term = **7**. [1 + 1]
8. $(3a - 2ab - 6) - (4a + 5ab - 7) = 3a - 2ab - 6 - 4a - 5ab + 7 =$ **$-a - 7ab + 1$** [1 for removing brackets, 1 for result]
9. Common factor = $3xy$. $6x^2y - 9xy^2 =$ **$3xy(2x - 3y)$**
10. $7x - 4x = 12 + 3$, so $3x = 15$, **$x = 5$** [1 + 1]

Section C — (3 marks each) **[12]**

11. $3x^2y \times (2x - 5y) = 6x^3y - 15x^2y^2$. [1½]
    Verify ($x = 2$, $y = -1$): LHS $= 3(4)(-1) \times (4 + 5) = -12 \times 9 = -108$. RHS $= 6(8)(-1) - 15(4)(1) = -48 - 60 = -108$. LHS = RHS. [1½]
    OR: $6x^2 - 18x + 2x - 5x^2 + 3x^2 =$ **$4x^2 - 16x$** [3]
12. $(4a + 9)(a - 6) = 4a^2 - 24a + 9a - 54 =$ **$4a^2 - 15a - 54$**. [1½]
    Verify ($a = 1$): LHS $= (4 + 9)(1 - 6) = 13 \times (-5) = -65$. RHS $= 4 - 15 - 54 = -65$. LHS = RHS. [1½]
    OR: $x(3x^2 - 2xy + 2y^2) - 2y(3x^2 - 2xy + 2y^2) = 3x^3 - 2x^2y + 2xy^2 - 6x^2y + 4xy^2 - 4y^3 =$ **$3x^3 - 8x^2y + 6xy^2 - 4y^3$** [3]
13. (i) Common factor $6ab$: **$6ab(2a - 3b)$** [1]
    (ii) $9a^2 - 30a + 25 = (3a)^2 - 2(3a)(5) + 5^2 =$ **$(3a - 5)^2$** [1]
    (iii) $2m^2 - 50 = 2(m^2 - 25) =$ **$2(m - 5)(m + 5)$** [1]
14. LCM of denominators = 6. Multiply both sides by 6: $2(x + 2) - 3(x - 1) = 1$. [1]
    $2x + 4 - 3x + 3 = 1$, so $-x + 7 = 1$, **$x = 6$**. [1] Check: $8/3 - 5/2 = 16/6 - 15/6 = 1/6$. [1]
    OR: Cross-multiply: $5(3x + 4) = -2(2 - 6x)$. [1] $15x + 20 = -4 + 12x$, so $3x = -24$, **$x = -8$**. [1]
    Check: $(-24 + 4)/(2 + 48) = -20/50 = -2/5$. [1]

Section D — Case-Based (4 marks) **[4]**

15. (i) Area = length × breadth $= (3x + 4)(2x - 1) = 6x^2 - 3x + 8x - 4 =$ **$6x^2 + 5x - 4$** m^2. [1]
    (ii) At $x = 5$: $6(25) + 5(5) - 4 = 150 + 25 - 4 =$ **$171\ m^2$**. [1]
    (iii) Perimeter = 2(length + breadth) $= 2(5x + 3) = 10x + 6$. [½] Equation: $10x + 6 = 56$. [½] $10x = 50$, **$x = 5$**. [1]

Note: Award full marks for any correct alternative method.